

An Integrative Review for Clinical Evaluation of Color Vision: The Right Test for the Right Disease

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Abstract

Purpose: To present an integrative review of the different color vision tests, their construction specificities, and their applications in ophthalmological and neurological diseases.

Methods: The literature was searched using the online databases such as PubMed, Scopus, Web of Science, and PsycInfo. The inclusion criteria included English studies, which focused on color measurement or evaluation for clinical diagnosis, involving group comparisons of congenital or acquired color deficiencies and healthy controls, participants over the age of 18 years, and published after 1970.

Results: We provide detailed descriptions of traditional and new computerized color vision tests including the test background, the prioritized level of processing, considerations regarding whether the test is more retinal or cortical tuned, if the task/skill measured is detection, discrimination or performance in color manipulation, and when to apply the respective tests.

Conclusion: The findings highlight the strong potential of color perception assessment in identifying early retinal changes and marking the progression of diseases, sometimes in subclinical conditions, in various ophthalmological and neurological conditions.

Keywords: Color blindness, Color perception, Diagnostic tests, Neurological diseases, Ophthalmic diseases

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INTRODUCTION

The retina is a signal transducer system, in which light energy information is transformed into electrochemical information about the spatial location of light, spatial contrast between adjacent areas of the retina, direction, and orientation of spatial visual stimuli, static and dynamic aspects of light, as well as the spectral characteristics of light.¹ All this information is processed in the retina and sent to the cortex to be the building blocks of visual perception.²

Our color perception is composed of three different perceptual dimensions that are representations of physical dimensions, such as (1) hue, which represents the tint, (2) saturation, which

corresponds to the level of white mixture in the color, and (3) brightness, which is related to the luminous intensity of the color.³ Thus, different color tests measure one or more of these dimensions, according to their composition, technology, and task to be performed. The anomaloscope, for example, is a measure of chromatic hue discrimination of the mixture of long (red) and green (medium) wavelengths compared to a medium-long wavelength light (yellow).⁴ The Cambridge Color Test (CCT), in turn, measures a chromaticity detection threshold, a color unit composed of hue and saturation, in relation to another reference chromaticity, usually the color white.^{5,6}

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Although ophthalmology historically uses visual acuity as a measure of the patient's visual function, other aspects of the visual experience, such as colors, textures, or depth, are often left without being clinically evaluated. Concerning color vision, only when there is a complaint about difficulty identifying colors is there a referral for a more specific assessment. For decades, color perception assessments have demonstrated potential for diagnosing subclinical visual reduction, such as in diabetes melitus,⁷⁻¹⁰ glaucoma,¹¹⁻¹⁶ and retinal intoxication by metallic mercury,¹⁷⁻²¹ in addition to knowing that color vision also contributes to the functional understanding of neurological diseases such as Parkinson's disease,²²⁻²⁵ multiple sclerosis with and without optic neuropathy,^{6,26,27} and Alzheimer's disease.²⁸⁻³²

In addition to the known contribution to the functional diagnosis of diseases and monitoring of therapeutic effects, there is little literature exploring the unique characteristics of color vision assessment instruments, which have different application purposes and diagnostic specificities.³³ Not knowing the different types of assessments, their methodologies, and their potential clinical contribution, little use is made of those important assessment tools in clinical practice.

An integrative review was considered since the aim of this research method is to gain an in-depth understanding of a given specification based on previous studies. Thus, we bring to this review article a succinct presentation of how much visual science has strengthened the understanding and representation of color vision, in the diversity of tests and their functional purposes, which have placed these visual tests at the level of those with high sensitivity diagnostic potential for diseases, retinal and cerebral. Based on robust scientific literature, we present which tests exist and for what purposes they should be applied, ending with the systematization of their potential to inform about the functional physiology of vision. Furthermore, we hypothesize for which groups of patients there is full evidence of diagnostic and functional benefits with assessment color vision clinic using specific tests reducing uncertainties of practical recommendations.

The color information

The eye is a signal-transducing organ, in which light energy is transformed into a neural information code with different functional characteristics and which is sent to various regions of the thalamus and brain stem.^{34,35} Among these characteristics is the differential processing of the spectral components of light, based on the different sensitivity to photon absorption of cone-type photoreceptors for different regions of the light spectrum.³⁶ In essence, photoreceptors are unable to signal the spectral origin of the light they capture within their spectrum, only the amount of quanta captured, which is known as the "Principle of Univariance".³⁷ The extraction of the spectral component from these different regions of the spectrum is then carried out by the second- and third-order neurons of the retina and, of main importance, are ganglion cells morphologically typified as dwarfs and bistratified ganglion cells [Figure 1].³⁸ In this way, the chromatic signal that leaves the retina is based

on a chromatic opponency signal spatially organized in center periphery.

In the visual cortex, in addition to the region marked by cytochrome oxidase in layers 2/3 of V1 (a region also known as "blobs"), chromatic information goes to V2, where it interacts with spatial information, being the probable origin of first order³⁹ chromatic contrast. Then, the chromatic and chromatic/spatial information goes to V4. In this cortical area, the fundamental characteristic is that the chromatic representation is bidirectionally balanced.⁴⁰ At least up to V1, there is a signal asymmetry in the S cone channel– (L cone + M cone). This implies that up to this stage of chromatic processing, we have receptive fields of S-ON and (L + M)-OFF polarity.³⁶ In V4, we have symmetrical responses both toward blue and toward yellow, suggesting an important modification of the chromatic signal, for another model that compensates responses from this chromatic pathway.⁴¹

This theoretical basis on the basic mechanisms of color vision is essential for understanding the differences between exams to evaluate color vision and also for understanding the different clinical pictures of color vision defects in the most varied ophthalmological and neurological diseases.

Color vision deficits

Congenital color vision deficiency (CVD) affects approximately 8% of males and 0.5% of females. Normal color vision requires the health of all three types of cone photoreceptors. Congenital CVD occurs when there is a loss of function in any type of cone photopigment.⁴² The most prevalent form is linked to the X chromosome and inherited as an X-linked recessive trait. It is characterized by the difficulty to discriminate between red and green color spectrum and is associated with the absence of either the long (L) or medium (M) wavelength-sensitive cone functions.⁴³

Acquired CVD, resulting from ocular or visual pathway disease, is known as acquired CVD. A recent review on the epidemiology of CVD suggested that acquired forms may affect between 5% and 15% of the population. However, it is essential to note that this claim relies primarily on level IV evidence, indicating expert opinion, rather than extensive surveys.⁴⁴

Verriest's classification of acquired CVD⁴⁵ is widely accepted and based on a retrospective analysis of 544 eyes from 476 patients who underwent various color vision tests. The classification includes:

- Type I acquired CVD: This involves a deficiency in the M-L mechanism ("red-green") with a shift in peak spectral sensitivity to shorter wavelengths
- Type II acquired CVD: This is an M-L mechanism deficiency but with relative preservation of the spectral sensitivity function
- Type III acquired CVD: This is an acquired S-mechanism ("blue-yellow") deficiency, possibly accompanied by a shift in peak spectral sensitivity to shorter wavelengths

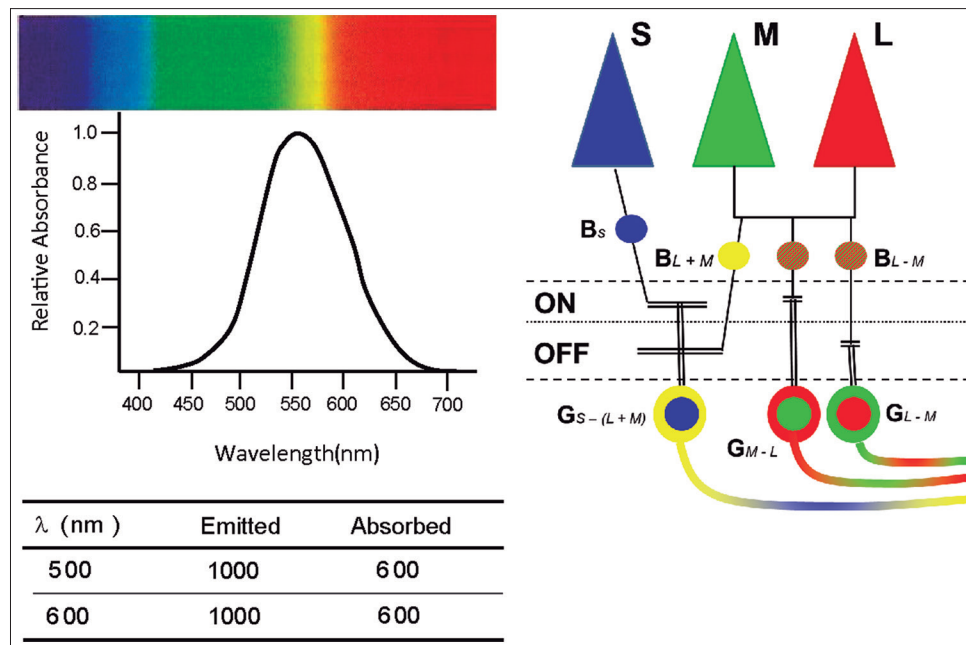


Figure 1: On the left we have a representation of the light spectrum between 400 nm and 700 nm, and a representation of the relative absorbance curve of a hypothetical photoreceptor, above a table that shows an example of metameric response between different wavelengths. In the right part we observe the post-receptoral arrangement in receptive fields of center-periphery antagonism, processing the information coming from the different cones L, M, and S, to create the physiological channels of chromatic opposition cone L– cone M and cone S– (cone L + cone M)

- Type IV Ill-defined/not classifiable: This category is for cases that do not clearly fit into the defined types or are difficult to classify.

Color measurement principles

Human color vision can discriminate about 32 million colors, considering changes in hue, brightness, and saturation. There are approximately 16 million colors in a high-fidelity color system.⁴⁶ Thus, a color image could contain more than a hundred colors even by dramatic quantization.³

The goals of color space quantization include (1) diminishing the dimension of the color histogram to minimize storage needs and enhance the speed of similarity measurement and (2) devising a relatively compact set of representative color bins from the entire color space to serve as a reference color set for histogram computation. To align with the human perception of similarity, where perceptually similar colors are anticipated to reside in the same quantized color bin, the quantization process should be executed in a perceptually uniform color space.⁴⁶ Many color models (spaces) have been designed to facilitate the color specification.⁴⁷ The key issue being considered in selection of a color space is the perceptual uniform property of that color space.

- The spectral primary system RGB. The RGB color space functions in a three-dimensional (3D) realm with axes for red (R), green (G), and blue (B), each ranging from 0 to 255 (256 levels).^{3,47} It is widely used and popular.
- Perceptual color space HLS or HSV. Another prevalent color space in computer graphics is HLS or HSV, which stands for hue (H), lightness (L), and saturation (S) or value (V).³ RGB coordinates can be smoothly converted

to HSV or HLS, despite the nonlinear nature of the conversion. These spaces are chosen for their practicality. In HLS or HSV, hue indicates the wavelength of the color, saturation measures purity, and value or lightness represents intensity.

- Munsell space and C.I.E. L/a/b/ space. Munsell and C.I.E. L/a/b/ spaces are also noteworthy. Munsell is a color-order space that organizes standard colors in a 3D setup of hue (H), value (V), and chroma (C), offering a perceptually uniform spacing of colors. C.I.E. L/a/b/ space quantifies the Munsell color space and is chosen for its perceptual uniformity, making it user-friendly for human perception.⁴⁸ When color adjustments are needed, components like hue, saturation, or lightness can be altered.
- Mathematical transformation to Munsell. The mathematical transformation to Munsell (MTM) space is preferred over traditional spaces like RGB because it exhibits superior perceptual uniformity. Experiments by Miyahara *et al.*⁴⁹ suggested that MTM provides more desirable HVC data than other transforming methods, including the conventional C.I.E. L/a/b/ method.

METHODS

The literature was searched using the online databases PubMed, Scopus, Web of Science, and PsycInfo. The indices of the proceedings of the International Research Group on Colour Vision Deficiencies were also checked via Springerlink. Search terms included “color(our) vision,” “color(our) perception,” or “chromatic sensitivity” coupled with the following: “test(ing),”

“evaluation”, and “measurement”. To be included in this review, articles had to meet the following criteria: Written in English; color measurement or evaluation for clinical diagnosis, group comparisons of congenital or acquired color deficiencies and healthy controls, and participants over age 18. We included empirical papers only published after 1970. Case studies were excluded.

The clinical tests

We present the different clinical exams available on the market for evaluating color vision, organized based on the prioritized level of processing and based on the type of task/skill being measured. There was no patient assessment; therefore, there was no need for informed consent for this study.

RESULTS

We briefly present the construction characteristics, procedures, and application for the main color vision tests found in the literature. There are a wide variety of color tests that are developed for different purposes, including tests to screen for dyschromatopsia. We did not include such tests in this review, considering that our purpose is to measure changes in color vision that occur due to diseases. Therefore, instruments such as Lantern color test^{50,51} and methods using eye-tracking⁵² were left out of this review.

Rayleigh equation

The matching principal evaluation is based on the Rayleigh equation and on metameric matching principles.³ The equation is obtained by mixing monochromatic 670 nm and 546 nm light to match with light of 589 nm.

Anomaloscope

Description

The instrument is considered the gold standard for diagnosing congenital color vision defects related to the L and M cones. It allows a quantitative assessment of two functional markers, the subjective equality point and the amplitude of equality, providing a quantitative classification of the color defect.⁴ However, it is a test that requires good collaboration and takes approximately 30 min per eye to complete the assessment.

Prioritized processing level

Retina, specifically L cones and M cones.

Task/skill measured

Detection of a chromatic difference between a yellow color from a monochromatic filter, with variability in the perception of yellow composed of the chromatic sum of green and red filters.

Application

Diagnosis of congenital color defect due to absence of opsin (protanopia and deuteranopia) or alteration of opsin structure (protanomaly and deuteranomaly). Quantitative measure of subjective equivalence point and equivalence range. Easy identification of the type of defect and precise quantification of its severity.

Pseudoisochromatics

Discrimination tests could be based in pseudoisochromatic principles.

Ishihara color plates (Kanehara Shuppan Co. Ltd., Tokyo, Japan)

Description

Presents 24 boards with stimuli defined by a pseudoisochromatic strategy, in which circles of different sizes and luminance with a chromatic difference between the figure and the background.⁵³ The first eight plates show whether color vision is normal. The other plates are used to determine the type of defect and its severity on a categorical scale of three levels, mild, moderate, and severe.

Prioritized processing level

Retina, specifically L cones and M cones, but requires symbolic identification of numbers, requiring processing of areas such as V4 and ventral visual areas such as Word Visual Form Area.

Task/ability measured

Perceptual sensitivity for discrimination between figure and background. The figure is made up of numbers that must be identified.

Application

Requires specific lighting. Diagnosis of congenital color defect due to absence of opsin (protanopia and deuteranopia) or alteration of opsin structure (protanomaly and deuteranomaly). Sensitivity is for moderate or higher defects and the defect severity level is in only 3 classification categories.

American Optical Hardy, Randy and Rittler (American Optical Company, New York, USA)

Description

Similar to the Ishihara test but detects congenital and acquired defects. It presents 20 boards with stimuli defined by a pseudoisochromatic strategy, in which circles of different sizes and luminance with a chromatic difference between the figure and the background.⁵⁴ The first six plates show whether color vision is normal. The remaining 14 plates are used to determine the type of protan, deutan, or tritan defect, and its severity on a 3-level categorical scale, mild, moderate, and severe.

Prioritized processing level

Retina, specifically L cones and M cones, but requires symbolic identification of geometric figures requiring processing of areas such as V3 and V4.

Task/ability measured

Perceptual sensitivity for discrimination between figure and background. The figure is made up of numbers that must be identified.

Application

Requires specific lighting. Diagnosis of congenital and acquired color defect. The sensitivity is for moderate or higher defects and the level of severity of the defect is in only 3 classification categories, classifying it as a screening test.

Cambridge Colour Test (Cambridge Research Systems, Rochester, UK)

Description

Computerized test that uses stimuli in a pseudoisochromatic format and the letter “C” of Landolt as the target.⁵⁵ Two testing protocols. Trivector screening protocol, as the name suggests, measures discrimination thresholds between figure and background in three axes of chromatic confusion: protan, deutan, and tritan. Complete Ellipses protocol measures MacAdam ellipses around specific points in the color space, allowing not only the classification of the defect but also the quantification of the level of severity of the loss, if any. An adaptive psychophysical routine that allows rapid testing measures the threshold. Trivector lasts approximately 5 min per eye and the complete Ellipses protocol, around 20 min per eye.

Prioritized processing level

Retina, L cones, M cones, and S cones, but requires symbolic identification of position, requiring processing of areas such as V3, V4, and V5.

Task/ability measured

Sensory sensitivity by measuring position discrimination thresholds of the Landolt “C” opening.

Application

Used for diagnosis of congenital and acquired color vision defects. The Trivector protocol, despite being a quick assessment, has good sensitivity and specificity. The Ellipses protocol allows a high degree of identification of defect types, as well as their classification in a quantitative and highly accurate manner. Excellent performance in monitoring the evolution of color vision during disease evolution, as well as measures of subsequent sequelae.

Tests based on ordering tasks

The subject’s task was to order the discs in a gradual progression in color considering a fix reference color.

Farnsworth-Munsell 100 hue (FMH100) (Pantone LLC, Carlstadt, USA)

Description

An instrument for evaluating performance in color ordering. Its 85 pieces with color surfaces based on the Munsell perceptual color space allow a quantitative and graphical assessment of the ability to order colors covering spectral and nonspectral colors with moderate saturation.⁵⁶ The difference between each color piece is above the threshold of two perceptual units (minimally perceptible difference). After observing the stones for 1 min, the stones are shuffled on a table with a dark surface. The measurement requires free time to make position adjustments between the stones. It lasts, on average, 20 min per eye.

Prioritized processing level: Visual cortex V2, V4 and V5 h, as this is a color surface ordering task.

Task/skill measured

Spatial ordering ability based only on the surface chromatic information of small parts.

Application

Requires specific lighting. It is primarily used to assess skills to deal with chromatic stimuli in spatial ordering tasks. It is therefore very useful for performance assessments with functional/occupational colors. The ability to diagnose color vision defects is low and has little sensitivity compared to the anomaloscope; it only detects moderate or higher defects.

Farnsworth 15 h (D15) (Pantone LLC, Carlstadt, USA)

Description

A reduced version of the Farnsworth-Munsell 100 hue, a screening instrument for evaluating color ordering performance. Contains 15 pieces with spectral and nonspectral colored surfaces.^{57,58} The difference between each color piece is above the threshold of six perceptual units. After observing the stones for 1 min, the caps are shuffled on a table with a dark surface. The measurement requires free time to make position adjustments between the caps. It lasts, on average, 15 min per eye.

Prioritize processing level

Visual cortex V2, V4, and V5 h, as this is a color surface ordering task.

Task/skill measured

Spatial ordering ability based only on the surface chromatic information of small parts.

Application

Requires specific lighting. It is very useful for performance assessments of functional/occupational colors. Used for functional and occupational purposes to screen for moderate and severe color vision defects. Used in conjunction with the Lanthony D15d for diagnosing acquired defects. Color vision detects diagnostic capacity is low and has little sensitivity compared to the anomaloscope, it only detects moderate or higher degree defects.

Lanthony (D15d) (Luneau, Paris, France)

Description

Version containing desaturated colors, with a lot of white, of the colors present in Farnsworth D15. Contains 15 pieces with spectral and nonspectral colored surfaces. The difference between each color piece is above the threshold of six perceptual units.⁵⁹ After observing the stones for 1 min, the stones are shuffled on a table with a dark surface. The measurement requires free time to make position adjustments between the stones. It lasts, on average, 15 min per eye.

Prioritized processing level

Visual cortex V2, V4, and V5 h, as this is a color surface ordering task.

Task/skill measured

Spatial ordering ability based only on the surface chromatic information of small parts.

Application

Requires specific lighting. They are used in conjunction with the Farnsworth D15 for diagnosing acquired defects. It has

great sensitivity to detect early acquired color vision defects, as well as to monitor the evolution of diseases.

These are color vision tests that are often used for clinical evaluation and are commercially available. There is a much wider variety of color vision measurements, based on other principles and specific assessments as Sahlgren's saturation test,⁶⁰ City University Test,⁶¹ Medmont C-100 color test,⁶² Berson plate test,⁶³ and color perimetry with short wavelength automated perimetry.⁶⁴ Electrophysiological measurements such as electroretinograms and visual evoked potentials can also assess color vision.⁶⁵ However, the need for specific monitors, calibration of emitters and gamma correction, the design of specific stimuli, and other technical specificities do not yet allow for their clinical availability. In Table 1, we present a summary of the tests described.

RESULTS PER DISEASES

Congenital dyschromatopsias

Congenital dyschromatopsia is a genetic condition in which alterations in the X chromosome lead to the absence of expression of one of the opsins, expressed in the L cone or the M cone. When amino acid exchange alterations occur instead of deletions, there is a shift in the sensitivity curve of one of these cones, leading to a reduction in the distance of the sensitivity peak between the cones and, therefore, a reduction in color discrimination.

The vast majority of color vision tests were developed to diagnose congenital dyschromatopsias. According to Birch studies, the gold standard method is the anomaloscope.^{43,66} However, in its absence, the diagnosis must be made by applying at least two methods. Conventionally, the Ishihara color test is the most widely used and, regardless of its version, containing more or less pseudoisochromatic plates, it has established itself as the main screening test for color vision defects. Something similar occurs with the American Optical HRR.^{54,67} Once the possibility of a color vision defect is detected in these tests, another test such as the Farnsworth D15 or Farnsworth 100 hue must be applied.⁶⁸ If confirmed, the diagnosis is established.

More recently, computerized methods have shown similar results in identifying congenital color vision defects.

Methods such as the CCT⁶⁹ and the Color Assessment and Diagnosis (CAD)⁷⁰ are the main examples.

Glaucoma

Few studies psychophysically analyze color vision in patients with glaucoma. Using FMH100, they observed no significant changes in the color vision of patients with high-pressure glaucoma, nor patients with open-angle glaucoma.¹³ They used the D15 and D15d tests in patients with glaucoma, classified as early defect and moderate defect, based on perimetric parameters. Both the groups already showed changes in color vision at D15d, suggesting that the test allows for very early detection of retinal damage.⁷¹

Thresholds drawing a MacAdam ellipse, similar to the procedure used by CCT, found changes for both the green-red axis and the blue-yellow axis.¹¹ However, the changes are earlier and more intense in the green-red axis when compared to the changes in the blue-yellow axis and precede the achromatic changes obtained in automated perimetry. The result was corroborated by the work of Miguel Castelo-Branco and collaborators,⁷² who found significant changes in the green-red and blue-yellow thresholds in the Trivector protocol and the Ellipses protocol of the CCT, in patients diagnosed with hypertensive glaucoma, but without changes perimeters.

Diabetic retinopathy

Studies evaluated patients with type 2 diabetes mellitus without clinical signs of diabetic retinopathy and absence of retinal angiographic changes using CCT and D15d.^{10,73} They found changes for both the green-red and blue-yellow defects in both methodologies. However, CCT showed greater sensitivity for the early detection of color defects when compared to D15d.¹⁰ Other studies corroborate our findings, suggesting D15d as the preferred test for clinical evaluation of patients with diabetes mellitus.⁵⁹

A recent study using Lanthony D15d shows that, compared with subjects with normal fasting glucose, the prevalence of acquired color vision impairment was higher among patients with type 2 diabetes without diabetic retinopathy, but not in subjects with impaired fasting glucose.⁷⁴ Thus, the acquired impairment of color vision may be attributed to another pathogenic mechanism associated with diabetic retinopathy.

Table 1: Summary of tests, their characteristics, and application purposes

Color vision test	Acronym	Processing level	Task	Application
Anomaloscope	-	Retina	Color matching	Diagnosis of congenital color blindness
Ishihara color plates	Ishihara	Retina	Figure-ground discrimination	Diagnosis of congenital color blindness
AO-HRR	AO-HRR	Retina	Figure-ground discrimination	Diagnosis of congenital and acquired color blindness
FMH100	FMH100	Cortical processing	Color spatial ordering	Ability in chromatic tasks; limited diagnosis
Farnsworth D15	D15	Cortical processing	Color spatial ordering	Ability in chromatic tasks; screening test for color blindness
Lanthony D15 desaturated	D15d	Cortical processing	Color spatial ordering	Ability in chromatic tasks; diagnostic test for acquired defects
CCT	CCT	Retina	Chromaticity discrimination threshold	Diagnosis of congenital and acquired color blindness

CCT: Cambridge Colour Test, AO-HRR: American Optical Hard, Randy and Rittler, FMH100: Farnsworth-Munsell 100 hue

Studies using FMH100 found differences only between controls and diabetics, however, no differences between patients with and without diabetic retinopathy⁷⁵ or in those patients in whom diabetic retinopathy is already established.^{75,76} The fact that the Farnsworth test uses a saturated color ordering task and a distance of two subjective units means it has low sensitivity to identify subtle defects.

Aged-related macular degeneration

Few studies on age-related macular degeneration (AMD) evaluate color function, which is at least curious, since the disease is macular, where the cones are concentrated. A study by Bowman, published in 1980,⁷⁷ used FMH100 and D15 in patients with AMD with few signs and in patients with advanced signs. The FMH100 ordering score was significantly higher, indicating worse efficiency in ordering colors, diffusely, without a specific axis of confusion, only in patients with advanced signs of the disease. The results for D15 were like those for FMH100.

A recent study used a methodology similar to CCT and found changes in color discrimination for both the green-red and blue-yellow axes in AMD patients, with mild but significantly worse results for the blue-yellow axis.⁷⁸

Multiple sclerosis

The ophthalmologic involvement of multiple sclerosis has a wide diversity of abnormalities. All parts of the visual sensory system, including the retina, optic nerve, chiasm, postchiasmatic pathways, and visual sensory and perceptual cortices and their connections to other brain areas, may be affected.⁷⁹ We carried out a study using CCT in patients with 6 months of disease remission, in which we identified that regardless of whether they had optic neuritis or not during the active phase of the disease, there is impairment of color vision.⁶ Our findings show that for both the groups of patients, the defects are diffuse, without a specific chromatic axis of involvement, and are corroborated by other studies.⁸⁰ Another study using the D15d test correlated diffuse changes in color vision with a reduction in the fiber layer of the optic nerve analyzed by optical coherence tomography.²⁷

Parkinson's disease

Changes in color vision have already been reported in patients with Parkinson's disease, with a reduction in retinal dopaminergic pathways and the visual pathway being suggested, its main etiology.^{81,82} D15d showed good sensitivity in detecting tritan color vision defects in patients with Parkinson's disease at the onset of motor symptoms.²³

The FMH100 test showed a moderate to low sensitivity for early detection of changes since only patients with mild cognitive impairment demonstrated significant changes in color ordering.²² Ishihara failed to detect color changes in 50% of patients with Parkinson's disease and essential tremor, considered by the authors to have low sensitivity for this population, and corroborated by another study that found no relationship between color vision and anatomical reduction

degenerative disease of the substantia nigra, due to the low sensitivity of the FMH100 test.⁸³

Neurotoxicity

The use of color vision tests is already well established as part of ophthalmological workup in cases of neurotoxicity. Several studies using different methodologies, such as CCT and D15, have successfully found reductions in chromatic discrimination and both tests have demonstrated great potential for diagnosis and therapeutic monitoring in this class of diseases. Both methodologies were sensitive to identifying diffuse changes in the color vision of workers occupationally exposed to mercury vapor,^{8,21} even in the absence of biological indicators such as metallic mercury in urine or hair. Amazonian riverside dwellers exposed to organic mercury through their diet have also had color vision losses identified using FM100 h and D15.⁸³⁻⁸⁶

A study on retinal neurotoxicity found diffuse color vision defects only in patients with more than 3 years of tamoxifen use and with an average of 36 g,⁸⁷ therefore not being sensitive to early detection of retinal effects. Another retinotoxic drug that generates a diffuse reduction in color discrimination is hydroxychloroquine. CCT and D15d have good sensitivity for detecting early defects, while AO-HRR and FMH100 only detect them in patients with greater changes.^{8,88,89}

Decreased color vision was identified after exposure to organic solvents through CCT, FMH100, and D15d.^{90,91} A diffuse loss of color vision was identified and exposure time to organic solvents was associated with losses in green-red color vision.

DISCUSSION

The different color vision tests provide a broad contribution to the functional assessment of different groups of patients, of which ophthalmological diseases and retinal changes generate negative impacts. Different color vision tests have demonstrated sensitivity, in addition to identifying functional changes, the possibility of being used as an instrument for monitoring the evolution of the disease and the treatments implemented.

Congenital CVD results from genetic disorders affecting the cone photopigments, genes regulating the expression of these photopigments, or genes encoding proteins involved in the phototransduction cascade, such as cone guanylate cyclase, GNAT2, and cone phosphodiesterase (PDE) subunits, including PDE6C and PDE6H.⁹² It can also arise from mutations in genes coding for the α - or β -subunits of cone cyclic guanosine monophosphate-gated cation channels.

To congenital dyschromatopsia the anomaloscope is the gold standard test, since it allows the identification of the type of defect, that is, whether there is an absence or alterations in the functions of the L cone or the M cone. The positive aspects include the precise quantification of the subjective chromatic equivalence point and the amplitude of chromatic tolerance.⁴³ Different degrees of normal color vision (normal,

weak, and deviated), as well as different degrees of anomalous trichromacy and dichromacy are performed by highly precise quantification. However, the measurement with the anomaloscope requires a lot of patience, as it is a highly precise instrument. There is a need to frequently readapt the eyes to white, which leads to an increase in the measurement time to around 45 min.

Acquired CVD has garnered significantly less attention compared to congenital CVD, which affects up to 8% of males and 0.5% of females.⁹³ Typically, acquired CVD impacts S-cone mediated discrimination, with recent epidemiological surveys indicating that cases involving the S-mechanism outnumber those affecting the M-L mechanism by at least a 2:1 ratio. Several hypotheses have been proposed to explain the apparent “vulnerability” of the S-mechanism. In certain disease states, Verriest’s type III defects may arise from various individual or combined mechanisms.

Evidence of S-cone-mediated color impairment is the main evidence in diseases such as glaucoma, type 2 diabetes mellitus, hydroxychloroquine and tamoxifen toxicity, and systemic heavy metal poisoning. Therefore, color tests such as Farnsworth D15, Lanthony D15d, and AO-HRR are the most sensitive for this type of diagnosis.^{10,85} Diseases such as AMD, retinal detachment, LHON, central serous retinopathy, dominant optic atrophy, and others affect color perception by both the S-cone and the L and M cones.⁶⁷ Therefore, tests such as CCT, CAD, and even D15 and D15d are the most indicated.

An overview of color testing

The best color vision test at the moment is the CCT, but its value is significantly higher than other tests. The use of the chromaticity diagram associated with an adaptive staircase psychophysical procedure, which modifies the presentation of the next stimulus based on the patient’s response history, allows for a dynamic variation of hue and saturation. Therefore, there is a great advantage over other tests in the high sensitivity diagnosis of both congenital and acquired defects. Its superiority in classifying and quantifying the defect is like that of the anomaloscope for congenital defects and proves to be superior for acquired defects when compared to D15d.

Considering a second class of tests with excellent sensitivity for acquired defects, whether due to optical or retinal diseases, are the Farnsworth D15 and Lanthony D15d ordering tests. The combined use of D15 and D15d allows for high-quality coverage for congenital and acquired defects. The Farnsworth D15 allows the identification of congenital color vision disorders, as well as the identification of defects acquired to an advanced degree of disease severity. The high level of color desaturation presented by D15d makes it very sensitive for detecting subtle changes in acquired defects and as demonstrated in some studies in our review, allowing the identification of defects even at a subclinical stage, mainly in glaucoma, type 2 diabetes, and neurotoxicity. retinal. The Farnsworth 100 hue test is a great test for evaluating functional

and occupational aspects of color vision, being the test of choice to be applied to pilots, drivers, and quality control workers, among others.¹³ Ishihara for congenital diagnosis and AO-HRR for congenital and acquired diseases have significant value for screening purposes and should be associated with another of these tests to confirm the diagnosis of color vision impairment.

The measurement of color pairing using the Rayleigh equation is still the gold standard for identifying and quantifying birth defects. The anomaloscope allows the identification of anomalous dyschromatopsia and trichromacy for genetic alterations of the L and M cones. The version of the anomaloscope for measuring S cone defects does not present the robustness of the version for red-green.

New tests and evaluation methodologies are developed regularly and at different levels and approaches to color processing. Minor examples include electroretinography work by silencing antagonistic responses and response separation by temporal frequency,⁵⁷⁻⁵⁹ by chromatic evoked visual potential,⁶⁰ and by psychophysical measurements using symmetric model color space to study color constancy⁶¹ and binocular color processing on surfaces.⁶²

We conclude this work by reinforcing the need for ophthalmologists, orthoptists, and other visual science professionals who deal with visual rehabilitation to employ color vision assessments in their clinical routine. As demonstrated, there is great potential for subclinical diagnosis of various retinal and neurological diseases, in addition to contributing to the monitoring of treatments in a functional and highly sensitive manner.

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Conflicts of interest

There are no conflicts of interest.

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